

## Prefix Sum

In Computer Science a prefix is a contiguous subarray that starts at every beginning of the array (index = 0) and extends upto some specific index  $i$ .

So in a prefix sum problem, the "prefix sum" simply means the running total or cumulative sum of all the elements from the start of the array upto the current position.

indices: 0 1 2 3

original Array: 5 2 8 1

Prefix Sum Array: 5 7 15 16

```
graph LR; subgraph Original_Array [original Array]; A0((5)); A1((2)); A2((8)); A3((1)); end; subgraph Prefix_Sum_Array [Prefix Sum Array]; P0((5)); P1((7)); P2((15)); P3((16)); end; A0 --> P0; A0 -- "+" --> P1; A1 -- "+" --> P2; A0 -- "+" --> P2; A1 -- "+" --> P3; A2 -- "+" --> P3; A3 -- "+" --> P3; A0 -- "+" --> P3;
```

$$\text{prefin}[i] = \text{prefin}[i-1] + \text{arr}[i]$$

The ultimate power of computing these prefixes is that it allows you to find the sum of any random middle section (subarray) of that array in a constant time  $O(1)$  without using any loop to re-add the numbers.

if you want the sum of elements from index  $L$  to  $R$  you take the prefix sum upto  $R$  and subtract the prefix sum just before  $L$ .

$$\text{Sum}(L, R) = \text{prefix}[R] - \text{prefix}[L-1]$$

| 0 | 1 | 2 | 3 |
|---|---|---|---|
| 5 | 2 | 8 | 1 |

(5) (7) (15)

$$\text{Sum}(1, 2) = 2 + 8 = 10$$

$$\text{prefix}(2) - \text{prefix}(0)$$

$$= 15 - 5 = 10$$

Postfix Sum (most commonly known as Suffix Sum)

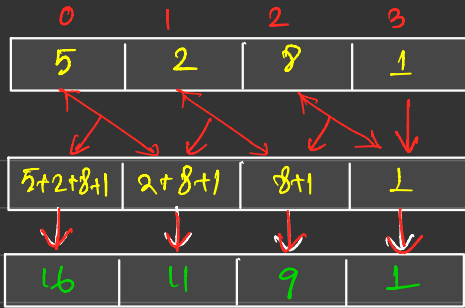
① The exact opposite of Prefix Sum.

Instead of adding elements from the beginning you add elements starting from the very end of the array (index  $n-1$ ), and work your way backward to the current position.

what is the suffix sum array :- A new array each index stores

the total sum of all elements from that index to the end of the array.

$$\text{suffix}[i] = \text{suffix}[i+1] + \text{arr}[i]$$



Actual Array



Suffix Array.  
or

Rank Array.